

Every Chemistry Formula

Class XII CBSE · Chemistry · Chapters 1–6

Every Chemistry Formula

Every formula the half-yearly can ask for, each with what its symbols mean, the unit it comes out in, and the one situation that tells you to reach for it. The cue is on the outside and the formula is hidden — so read down the page and say each one before you open it.

- Formulas: 50
- Recall rules: 18
- Must be instant: 40
- Paper: 70 marks

How to use this

● means you should be able to say it before the question finishes loading. ○ means a few seconds of thought is fine — these are the ones you derive or reconstruct rather than recall.

Work down a chapter with everything closed. Say the formula out loud, *then* tap to reveal. A formula you can only recognise is not a formula you have.

Units are not decoration. Most lost marks in Solutions and Electrochemistry are unit slips — molality is per **kilogram of solvent**, molarity per **litre of solution**, and κ in S cm^{-1} with c in mol L^{-1} is what makes the 1000 appear in molar conductivity.

Recognise index

Every cue on one screen. Read the cue, say the formula, tap to check.

1 · Solutions — 15 marks

- C1.1 Concentration by mass, volume, or parts per million
- C1.2 Mole fraction of a component
- C1.3 Molarity, from mass or from moles
- C1.4 Molality, and why it is the one used for colligative properties
- C1.5 A gas dissolving in a liquid under pressure
- C1.6 Vapour pressure of a mixture of two volatile liquids
- C1.7 Vapour pressure when the solute is non-volatile
- C1.8 Relative lowering of vapour pressure
- C1.9 Boiling point raised by a dissolved solute
- C1.10 Freezing point lowered by a dissolved solute
- C1.11 Osmotic pressure across a semipermeable membrane
- C1.12 Molar mass from any one colligative property

- C1.13 The measured effect is bigger or smaller than predicted
- C1.14 Degree of dissociation or association from i

2 · Electrochemistry — 14 marks

- C2.1 Standard cell potential from two electrode potentials
- C2.2 Cell potential at non-standard concentrations
- C2.3 Nernst for a single electrode
- C2.4 Free energy change of a cell reaction
- C2.5 Equilibrium constant from cell potential
- C2.6 Conductance, resistance and the cell constant
- C2.7 Conductivity from measured resistance
- C2.8 Molar conductivity from conductivity
- C2.9 Limiting molar conductivity of an electrolyte from its ions
- C2.10 Degree of dissociation of a weak electrolyte
- C2.11 Dissociation constant of a weak acid from conductivity
- C2.12 Mass deposited during electrolysis
- C2.13 Moles of electrons passed

3 · Chemical Kinetics — 13 marks

- C3.1 Rate of a reaction from a concentration change
- C3.2 Rate written for a balanced equation with coefficients
- C3.3 Rate law and the order of a reaction
- C3.4 Units of the rate constant for any order
- C3.5 Zero order — concentration against time
- C3.6 Zero order half-life
- C3.7 First order rate constant from concentrations
- C3.8 First order half-life
- C3.9 First order reaction in the gas phase, measured by total pressure
- C3.10 Rate constant against temperature
- C3.11 Activation energy from a straight-line plot
- C3.12 Activation energy from two temperatures

4 · d and f Block — 11 marks

- C4.1 Magnetic moment from unpaired electrons, and back again
- C4.2 General electronic configuration of a block

5 · Coordination Compounds — 11 marks

- C5.1 Oxidation number of the metal inside a complex
- C5.2 Coordination number when a ligand grips more than once

- C5.3 Tetrahedral splitting compared with octahedral
- C5.4 Crystal field stabilisation energy of an octahedral complex
- C5.5 High spin or low spin — deciding which
- C5.6 Number of geometrical isomers of a complex

6 · Haloalkanes and Haloarenes — 6 marks

- C6.1 Rate law that identifies the substitution mechanism
- C6.2 Degree of unsaturation from a molecular formula

Constants — worth the same marks as the formulas

- K Every constant this paper can hand you, with its unit

CH 1 Solutions — 14 formulas · 15 marks

● C1.1 Concentration by mass, by volume, or in parts per million

$$\% w/w = \frac{w_B}{w_A + w_B} \times 100 \quad \% v/v = \frac{V_B}{V_{\text{soln}}} \times 100$$

$$\% w/v = \frac{w_B}{V_{\text{soln in mL}}} \times 100 \quad \text{ppm} = \frac{w_B}{w_{\text{soln}}} \times 10^6$$

Symbol	Meaning	Unit
w_B	mass of solute	g
w_A	mass of solvent	g
V_{soln}	volume of solution	mL or L
all four	the concentration itself	dimensionless

Use it when:

The question gives masses or volumes and no moles anywhere. ppm is for trace quantities — pollutants, dissolved gases, hardness of water.

Trap:

$\% w/w$ divides by the mass of the **solution**, not the solvent. Adding 10 g of solute to 90 g of water gives 10%, not 11.1%.

● C1.2 Mole fraction of a component

$$x_A = \frac{n_A}{n_A + n_B} \quad x_A + x_B = 1$$

Symbol	Meaning	Unit
x_A	mole fraction of the solvent	dimensionless
n_A	moles of solvent, $= w_A/M_A$	mol
n_B	moles of solute	mol

Use it when:

Raoult's law or Henry's law is involved — both are written in mole fractions, never in molarity.

Trap:

For water as solvent, $M_A = 18$, so n_A is large and x_B is tiny. Do not round x_A to 1 before subtracting — you will get $x_B = 0$.

● **C1.3 Molarity, from moles or straight from mass**

$$M = \frac{n_B}{V_{\text{soln in L}}} = \frac{w_B \times 1000}{M_B \times V_{\text{soln in mL}}}$$

Symbol	Meaning	Unit
M	molarity	mol L ⁻¹
n_B	moles of solute	mol
M_B	molar mass of solute	g mol ⁻¹
V	volume of solution	L (or mL)

Use it when:

Volume of solution is given or asked. Also the concentration used in osmotic pressure, $\pi = CRT$.

Trap:

Molarity **changes with temperature** because volume does. Molality does not. If a question mentions heating or cooling and then asks which concentration term is unaffected, the answer is molality.

● **C1.4 Molality — and the unit that makes it the colligative one**

$$m = \frac{n_B}{w_A \text{ in kg}} = \frac{w_B \times 1000}{M_B \times w_A \text{ in g}}$$

Symbol	Meaning	Unit
m	molality	mol kg^{-1}
w_A	mass of solvent only	kg (or g)
M_B	molar mass of solute	g mol^{-1}

Use it when:

Anything colligative: ΔT_b , ΔT_f . Both are written in molality precisely because it is temperature-independent.

Trap:

Per kg of **solvent**, not of solution. The 1000 in the second form converts grams of solvent to kilograms — it is not the same 1000 as in molarity, where it converts mL of solution to L.

● C1.5 A gas dissolving in a liquid under pressure

$$p = K_H x$$

Symbol	Meaning	Unit
p	partial pressure of the gas above the solution	bar (or kbar)
x	mole fraction of the gas in solution	dimensionless
K_H	Henry's law constant	bar (or kbar)

Use it when:

A gas is being dissolved — soda bottles, deep-sea diving and the bends, oxygen at altitude, aquatic life in warm water.

Trap:

K_H goes **up** with temperature, so solubility goes **down** — which is why warm water holds less dissolved oxygen. A higher K_H means a *less* soluble gas.

● C1.6 Vapour pressure of a mixture of two volatile liquids

$$p_{\text{total}} = p_A^\circ x_A + p_B^\circ x_B = p_B^\circ + (p_A^\circ - p_B^\circ) x_A$$

Symbol	Meaning	Unit
p_A°	vapour pressure of pure A	bar or mm Hg
x_A	mole fraction of A in the liquid	dimensionless
y_A	mole fraction of A in the vapour , $= p_A/p_{\text{total}}$	dimensionless

Use it when:

Both components evaporate. The second form is the equation of a straight line in x_A — that is the plot they ask you to sketch.

Trap:

Vapour composition y_A is not the same as liquid composition x_A . The vapour is always richer in the more volatile component.

● **c1.7 Vapour pressure when the solute is non-volatile**

$$p_{\text{soln}} = x_A p_A^\circ$$

Symbol	Meaning	Unit
p_{soln}	vapour pressure of the solution	bar or mm Hg
x_A	mole fraction of the solvent	dimensionless
p_A°	vapour pressure of pure solvent	bar or mm Hg

Use it when:

A solid, non-volatile solute (sugar, urea, a salt) is dissolved. Only the solvent contributes vapour.

Trap:

It is x of the **solvent**, not the solute. Using x_B here is the single commonest slip in this chapter.

● **c1.8 Relative lowering of vapour pressure**

$$\frac{p_A^\circ - p_{\text{soln}}}{p_A^\circ} = x_B = \frac{n_B}{n_A + n_B} \approx \frac{w_B M_A}{M_B w_A}$$

Symbol	Meaning	Unit
x_B	mole fraction of solute	dimensionless
M_A	molar mass of solvent	g mol^{-1}
M_B	molar mass of solute — usually what is asked	g mol^{-1}

Use it when:

Two vapour pressures are given (pure solvent and solution) and a molar mass is wanted.

Trap:

The final approximation assumes $n_B \ll n_A$ — valid for a dilute solution only. If the question says concentrated, use the exact form.

● C1.9 Boiling point raised by a dissolved solute

$$\Delta T_b = K_b m \quad \Delta T_b = T_b - T_b^\circ$$

Symbol	Meaning	Unit
ΔT_b	elevation of boiling point	K
K_b	molal elevation constant (ebullioscopic constant)	K kg mol ⁻¹
m	molality	mol kg ⁻¹

Use it when:

A boiling point shifts. For water $K_b = 0.52 \text{ K kg mol}^{-1}$.

Trap:

K_b is a property of the **solvent** alone — it never depends on which solute you dissolved.

● C1.10 Freezing point lowered by a dissolved solute

$$\Delta T_f = K_f m \quad \Delta T_f = T_f^\circ - T_f$$

Symbol	Meaning	Unit
ΔT_f	depression of freezing point	K
K_f	molal depression constant (cryoscopic constant)	K kg mol ⁻¹
m	molality	mol kg ⁻¹

Use it when:

A freezing point shifts — antifreeze in a radiator, salt on an icy road. For water $K_f = 1.86 \text{ K kg mol}^{-1}$.

Trap:

Note the order of subtraction is reversed from ΔT_b : freezing point goes **down**, so it is pure minus solution. Both ΔT values come out positive.

● C1.11 Osmotic pressure across a semipermeable membrane

$$\pi = CRT = \frac{n_B}{V}RT \quad \pi V = n_B RT$$

Symbol	Meaning	Unit
π	osmotic pressure	bar or atm
C	molarity	mol L ⁻¹
R	gas constant	0.0821 L atm K ⁻¹ mol ⁻¹
T	absolute temperature	K

Use it when:

A membrane, a molar mass of a **protein or polymer**, or the word isotonic/hypertonic appears. This is the method of choice for large molar masses because π is measurably large even when ΔT_f would be tiny.

Trap:

Uses molarity, not molality — the only colligative property that does. Match the R value to the pressure unit you want.

● **C1.12 Molar mass from any one colligative property**

$$M_B = \frac{K_b w_B \times 1000}{\Delta T_b w_A} \quad M_B = \frac{K_f w_B \times 1000}{\Delta T_f w_A} \quad M_B = \frac{w_B RT}{\pi V}$$

Symbol	Meaning	Unit
M_B	molar mass of the solute	g mol ⁻¹
w_B	mass of solute	g
w_A	mass of solvent	g
V	volume of solution	L

Use it when:

"Calculate the molar mass of the solute" — the standard 3-marker. All three are just C1.9–C1.11 rearranged; do not memorise them separately.

Trap:

The 1000 is there because w_A is in grams while molality needs kilograms. Drop it and your answer is out by exactly a factor of 1000.

● **C1.13 When the measured effect is bigger or smaller than predicted**

$$i = \frac{\text{observed colligative property}}{\text{calculated colligative property}} = \frac{\text{normal molar mass}}{\text{abnormal molar mass}}$$

$$\Delta T_b = iK_b m \quad \Delta T_f = iK_f m \quad \pi = iCRT$$

Symbol	Meaning	Unit
i	van't Hoff factor	dimensionless
$i > 1$	solute dissociates — e.g. NaCl, $i \rightarrow 2$	dimensionless
$i < 1$	solute associates — e.g. benzoic acid in benzene, $i \rightarrow 0.5$	dimensionless

Use it when:

The solute is ionic, or the question uses the word "abnormal", or the observed ΔT_f does not match the calculated one.

Trap:

Molar mass and i move **opposite** ways. Dissociation ($i > 1$) makes the measured molar mass come out *lower* than the true value.

○ C1.14 Degree of dissociation or association from i

$$\alpha_{\text{dissociation}} = \frac{i - 1}{n - 1} \quad \alpha_{\text{association}} = \frac{1 - i}{1 - \frac{1}{n}}$$

Symbol	Meaning	Unit
α	degree of dissociation or association	dimensionless
n	number of particles one formula unit gives (or combines into)	dimensionless
i	van't Hoff factor	dimensionless

Use it when:

i has been found and the percentage dissociation is asked. For K_2SO_4 , $n = 3$; for a dimerising acid, $n = 2$.

Trap:

Read n off the actual dissociation equation, not the number of atoms. $\text{K}_2\text{SO}_4 \rightarrow 2\text{K}^+ + \text{SO}_4^{2-}$ gives three particles, so $n = 3$.

CH 2 Electrochemistry — 13 formulas · 14 marks

● C2.1 Standard cell potential from two electrode potentials

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} = E_{\text{right}}^{\circ} - E_{\text{left}}^{\circ}$$

Symbol	Meaning	Unit
E_{cell}°	standard cell potential	V
E°	standard reduction potential of each electrode	V

Use it when:

Two half-cells and their E° values are given. A positive E°_{cell} means the reaction is spontaneous as written. Daniell cell: 1.1 V.

Trap:

Both values in the table are **reduction** potentials — subtract, never add, and never flip the sign of the anode value and then also subtract.

● **c2.2 Cell potential when concentrations are not 1 M**

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{2.303 RT}{nF} \log Q \xrightarrow{298 \text{ K}} E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.059}{n} \log Q$$

Symbol	Meaning	Unit
n	moles of electrons in the balanced cell reaction	dimensionless
Q	reaction quotient, products over reactants	dimensionless
F	Faraday constant	96487 C mol ⁻¹
R	gas constant	8.314 J K ⁻¹ mol ⁻¹

Use it when:

Any concentration is given that is not 1 M, or the question says "calculate the emf of the cell".

Trap:

NCERT uses **0.059**, not 0.0591. Pure solids and liquids never appear in Q . Get n from the balanced equation, not from the charge on one ion.

○ **c2.3 Nernst equation for a single electrode**

$$E_{(\text{M}^{n+}/\text{M})} = E^\circ_{(\text{M}^{n+}/\text{M})} - \frac{0.059}{n} \log \frac{1}{[\text{M}^{n+}]}$$

Symbol	Meaning	Unit
$[\text{M}^{n+}]$	concentration of the metal ion	mol L ⁻¹
n	charge on the metal ion	dimensionless
E	electrode potential	V

Use it when:

Only one half-cell is being asked about, or a concentration cell is given where both electrodes are the same metal at different concentrations.

Trap:

The solid metal is taken as unit activity, so it is the 1 in the numerator. For a concentration cell $E_{\text{cell}}^{\circ} = 0$ and the whole emf comes from the log term.

● C2.4 Free energy change of a cell reaction

$$\Delta_r G = -nFE_{\text{cell}} \quad \Delta_r G^{\circ} = -nFE_{\text{cell}}^{\circ}$$

Symbol	Meaning	Unit
$\Delta_r G$	Gibbs energy change of the cell reaction	J mol ⁻¹
n	moles of electrons transferred	dimensionless
F	Faraday constant	96487 C mol ⁻¹

Use it when:

The question asks for ΔG , for maximum work obtainable, or whether a reaction is feasible.

Trap:

Answer comes out in joules, not kilojoules — divide by 1000 if the options are in kJ. The minus sign is part of the formula: a positive emf gives a negative ΔG .

○ C2.5 Equilibrium constant from cell potential

$$\Delta_r G^{\circ} = -RT \ln K_c \quad \implies \quad E_{\text{cell}}^{\circ} = \frac{0.059}{n} \log K_c$$

Symbol	Meaning	Unit
K_c	equilibrium constant of the cell reaction	dimensionless
E_{cell}°	standard cell potential	V
n	moles of electrons	dimensionless

Use it when:

K_c is asked from E° , or vice versa. It is just the Nernst equation with $E_{\text{cell}} = 0$ and $Q = K_c$ — at equilibrium the cell is dead.

Trap:

Note this is +, not -: setting $E_{\text{cell}} = 0$ moves the log term across the equals sign.

● C2.6 Conductance, resistance and the cell constant

$$G = \frac{1}{R} \quad G^* = \frac{l}{A} \quad R = \rho \frac{l}{A}$$

Symbol	Meaning	Unit
G	conductance	S (siemens)
R	resistance	Ω
G^*	cell constant	cm^{-1} or m^{-1}
ρ	resistivity	$\Omega \text{ cm}$

Use it when:

A conductivity cell is described by its electrode area and separation, or the cell constant is given directly.

Trap:

$1 \text{ S} = 1 \Omega^{-1}$, also written mho. The cell constant has units of **inverse** length because it is $\text{length} \div \text{area}$.

● **C2.7 Conductivity from a measured resistance**

$$\kappa = \frac{1}{\rho} = G \times G^* = \frac{1}{R} \times \frac{l}{A}$$

Symbol	Meaning	Unit
κ	conductivity (specific conductance)	S cm^{-1} or S m^{-1}
G^*	cell constant	cm^{-1}
R	measured resistance	Ω

Use it when:

The question gives a resistance from a conductivity cell. κ is the conductance of a 1 cm cube of solution.

Trap:

κ **decreases** on dilution (fewer ions per unit volume) while molar conductivity **increases**. Being asked to explain that opposite behaviour is a standing 2-marker.

● **C2.8 Molar conductivity from conductivity**

$$\Lambda_m = \frac{\kappa}{c} = \frac{\kappa \times 1000}{c \text{ in mol L}^{-1}}$$

Symbol	Meaning	Unit
Λ_m	molar conductivity	$\text{S cm}^2 \text{ mol}^{-1}$
κ	conductivity	S cm^{-1}
c	concentration	mol L^{-1}

Use it when:

Converting a measured κ into the quantity you actually compare between electrolytes.

Trap:

The 1000 exists only because κ is per **cm³** while c is per **litre**. In SI (κ in S m⁻¹, c in mol m⁻³) there is no 1000 and the unit is S m² mol⁻¹.

● C2.9 Limiting molar conductivity built from its ions

$$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$$

Symbol	Meaning	Unit
Λ_m°	limiting molar conductivity (infinite dilution)	S cm ² mol ⁻¹
$\lambda_\pm^\circ$	limiting molar conductivity of each ion	S cm ² mol ⁻¹
$\nu_\pm$	how many of that ion one formula unit gives	dimensionless

Use it when:

Λ_m° of a **weak** electrolyte is wanted — it cannot be measured by extrapolation, so it is assembled from strong electrolytes. This is **Kohlrausch's law of independent migration of ions**.

Trap:

Multiply by ν . For CaCl₂, $\Lambda_m^\circ = \lambda_{\text{Ca}^{2+}}^\circ + 2\lambda_{\text{Cl}^-}^\circ$ — the 2 is easy to drop.

● C2.10 Degree of dissociation of a weak electrolyte

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$$

Symbol	Meaning	Unit
α	degree of dissociation	dimensionless
Λ_m	molar conductivity at concentration c	S cm ² mol ⁻¹
Λ_m°	limiting molar conductivity	S cm ² mol ⁻¹

Use it when:

A weak acid or base is given with both conductivity values.

Trap:

Only meaningful for a **weak** electrolyte. A strong electrolyte is fully dissociated; its Λ_m rises with dilution because ion-ion interference falls, not because more of it dissociates.

○ C2.11 Dissociation constant of a weak acid from conductivity

$$K_a = \frac{c\alpha^2}{1-\alpha} = \frac{c\Lambda_m^2}{\Lambda_m^\circ(\Lambda_m^\circ - \Lambda_m)}$$

Symbol	Meaning	Unit
K_a	dissociation constant	mol L ⁻¹
c	concentration of the acid	mol L ⁻¹
α	degree of dissociation	dimensionless

Use it when:

Both Λ_m and Λ_m° are given for a weak acid and K_a or pK_a is asked.

Trap:

Do not approximate $1 - \alpha \approx 1$ unless α is genuinely small (below about 0.05).

● C2.12 Mass deposited or liberated during electrolysis

$$Q = It \quad w = ZIt \quad Z = \frac{M}{nF}$$

Symbol	Meaning	Unit
Q	charge passed	C
I	current	A
t	time	s
w	mass deposited	g
Z	electrochemical equivalent	g C ⁻¹

Use it when:

A current runs for a time and a mass is asked. Faraday's first law of electrolysis.

Trap:

Time must be in **seconds**. A question quoting minutes or hours is testing exactly this.

● C2.13 Moles of electrons passed, and how much substance that gives

$$n_{e^-} = \frac{Q}{F} = \frac{It}{96487} \quad \text{moles of product} = \frac{n_{e^-}}{n}$$

Symbol	Meaning	Unit
n_{e^-}	moles of electrons	mol
F	charge on one mole of electrons	96487 C mol ⁻¹
n	electrons needed per ion, from the half-equation	dimensionless

Use it when:

Several cells are in series, or a gas volume at STP is asked. Faraday's second law: the same charge deposits amounts in the ratio of their equivalent masses.

Trap:

Get n from the half-reaction. $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$ needs 2 F per mole; Al^{3+} needs 3 F.

CH 3 Chemical Kinetics — 12 formulas · 13 marks

● C3.1 Rate of a reaction from a concentration change

$$r_{\text{av}} = -\frac{\Delta[R]}{\Delta t} = +\frac{\Delta[P]}{\Delta t} \quad r_{\text{inst}} = -\frac{d[R]}{dt}$$

Symbol	Meaning	Unit
$[R]$	concentration of a reactant	mol L ⁻¹
t	time	s
r	rate of reaction	mol L ⁻¹ s ⁻¹

Use it when:

A table of concentration against time is given. Instantaneous rate is the slope of the tangent to that curve.

Trap:

The minus sign exists so the rate comes out positive — reactant concentration is falling. Never report a negative rate.

● C3.2 Rate written for an equation that has coefficients



$$r = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = +\frac{1}{c} \frac{d[C]}{dt} = +\frac{1}{d} \frac{d[D]}{dt}$$

Symbol	Meaning	Unit
a, b, c, d	stoichiometric coefficients	dimensionless
r	the one unique rate of reaction	$\text{mol L}^{-1} \text{s}^{-1}$

Use it when:

The question gives the rate of disappearance of one species and asks for another.

$2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$ makes NO_2 four times as fast as it makes O_2 .

Trap:

You **divide** by the coefficient. Multiplying is the reflex error.

● **C3.3 Rate law, and the order of a reaction**

$$r = k [A]^x [B]^y \quad \text{order} = x + y$$

Symbol	Meaning	Unit
k	rate constant	depends on order — see C3.4
x, y	order with respect to A and B	dimensionless
order	overall order	dimensionless

Use it when:

A table of initial rates against initial concentrations is given. Double one concentration, see what the rate does: unchanged $\rightarrow$ order 0, doubles $\rightarrow$ 1, quadruples $\rightarrow$ 2.

Trap:

Order is found by **experiment** and can be zero or fractional. Molecularity comes from the mechanism, is a whole number, and is never zero. A **pseudo first order reaction** is second order on paper but first order in practice because one reactant is in huge excess.

● **C3.4 Units of the rate constant for any order**

$$[k] = \text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$$

Symbol	Meaning	Unit
$n = 0$	zero order	$\text{mol L}^{-1} \text{s}^{-1}$
$n = 1$	first order	s^{-1}
$n = 2$	second order	$\text{L mol}^{-1} \text{s}^{-1}$
n	overall order	dimensionless

Use it when:

The order is asked and only the units of k are given — this runs backwards perfectly, and it is a favourite one-marker.

Trap:

First order k has no concentration unit at all, which is why a first-order half-life does not depend on concentration.

● **C3.5 Zero order — concentration against time**

$$[R] = -kt + [R]_0 \quad k = \frac{[R]_0 - [R]}{t}$$

Symbol	Meaning	Unit
$[R]_0$	initial concentration	mol L^{-1}
$[R]$	concentration at time t	mol L^{-1}
k	zero order rate constant	$\text{mol L}^{-1} \text{ s}^{-1}$

Use it when:

The plot of $[R]$ against t is a straight line of slope $-k$. Real examples: decomposition of NH_3 on hot platinum, of HI on gold.

Trap:

Zero order means the rate is independent of concentration — usually because a catalyst surface is saturated, not because nothing is happening.

● **C3.6 Zero order half-life**

$$t_{1/2} = \frac{[R]_0}{2k}$$

Symbol	Meaning	Unit
$t_{1/2}$	half-life	s
$[R]_0$	initial concentration	mol L^{-1}
k	rate constant	$\text{mol L}^{-1} \text{ s}^{-1}$

Use it when:

A zero-order reaction's half-life is asked, or the question tests that $t_{1/2} \propto [R]_0$ here.

Trap:

This one **does** depend on starting concentration — the exact opposite of first order. Getting the two the wrong way round is the standing trap of this chapter.

● C3.7 First order rate constant from concentrations

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]} \quad \log[R] = -\frac{k}{2.303}t + \log[R]_0$$

Symbol	Meaning	Unit
k	first order rate constant	s^{-1}
$[R]_0$	initial concentration	mol L^{-1}
$[R]$	concentration at time t	mol L^{-1}

Use it when:

The commonest numerical in the chapter. The second form says a plot of $\log[R]$ against t is a straight line of slope $-k/2.303$.

Trap:

2.303 converts $\ln$ to $\log_{10}$. If you use $\ln$, drop the 2.303 — using both is a factor-of-2.3 error.

● C3.8 First order half-life

$$t_{1/2} = \frac{0.693}{k}$$

Symbol	Meaning	Unit
$t_{1/2}$	half-life	s
k	first order rate constant	s^{-1}
0.693	$\ln 2$	dimensionless

Use it when:

Any half-life question on a first-order reaction, including all radioactive decay.

Trap:

Independent of starting concentration. Being asked to prove that is a standing 2-marker — see the derivations page.

○ C3.9 First order gas-phase reaction followed by total pressure

$$k = \frac{2.303}{t} \log \frac{p_i}{2p_i - p_t}$$

Symbol	Meaning	Unit
p_i	initial pressure of the reactant	bar or atm
p_t	total pressure at time t	bar or atm
k	rate constant	s^{-1}

Use it when:

A gas decomposes into two gas molecules and the data is a table of total pressure against time — e.g. azoisopropane, or N_2O_5 decomposition.

Trap:

The $2p_i - p_t$ form holds only when one mole of gas gives two moles of gas. For a different stoichiometry, rebuild the pressure table from scratch.

● C3.10 Rate constant against temperature

$$k = A e^{-E_a/RT}$$

Symbol	Meaning	Unit
A	frequency (pre-exponential) factor	same as k
E_a	activation energy	J mol^{-1}
R	gas constant	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
T	absolute temperature	K

Use it when:

Temperature enters the question at all. The Arrhenius equation.

Trap:

$e^{-E_a/RT}$ is the **fraction of molecules with energy at least E_a** — quoting that meaning is often the mark.

○ C3.11 Activation energy from a straight-line plot

$$\log k = \log A - \frac{E_a}{2.303 RT}$$

Symbol	Meaning	Unit
slope	$-E_a/2.303R$	K
intercept	$\log A$	dimensionless
x -axis	$1/T$	K^{-1}

Use it when:

A graph of $\log k$ against $1/T$ is given, or asked to be sketched.

Trap:

The slope is **negative** and E_a is positive — remember to take the magnitude when you convert slope to E_a .

● **C3.12 Activation energy from two temperatures**

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right] = \frac{E_a}{2.303 R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

Symbol	Meaning	Unit
k_1, k_2	rate constants at T_1 and T_2	same units
E_a	activation energy	J mol ⁻¹
T_1, T_2	absolute temperatures	K

Use it when:

Two rate constants at two temperatures are given. Also handles "the rate doubles when temperature rises by 10 K" — set $k_2/k_1 = 2$.

Trap:

Inside the bracket it is $1/T_1 - 1/T_2$ — the **smaller** temperature first. Reversing it flips the sign of E_a .

CH 4 d and f Block Elements — 2 formulas + rules · 11 marks

This chapter is mostly trends and reasons rather than calculation. Two things are computable; everything else is recall, collected in the rules table below.

● **C4.1 Magnetic moment from unpaired electrons — and back again**

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

Symbol	Meaning	Unit
μ	spin-only magnetic moment	BM (Bohr magneton)
n	number of unpaired electrons	dimensionless
$n = 1, 2, 3$	gives $\mu = 1.73, 2.83, 3.87$	BM
$n = 4, 5$	gives $\mu = 4.90, 5.92$	BM

Use it when:

Either direction: count unpaired electrons from the configuration and get μ , or read μ off

the question and work back to n . Same formula serves Chapter 5.

Trap:

n is unpaired electrons in the **ion**, not the atom. For a transition metal ion, remove the $4s$ electrons first, then the $3d$.

○ C4.2 General electronic configuration of each block

d-block: $(n-1)d^{1-10}ns^{1-2}$ lanthanoid: $4f^{1-14}5d^{0-1}6s^2$ actinoid: $5f^{1-14}6d^{0-1}7s^2$

Symbol	Meaning	Unit
n	period number	dimensionless
Cr	exception: $3d^54s^1$, half-filled is stabler	dimensionless
Cu	exception: $3d^{10}4s^1$, fully-filled is stabler	dimensionless

Use it when:

Writing a configuration, or explaining why Zn, Cd and Hg are not really transition metals — their d subshell is full in both atom and ion.

Trap:

Electrons fill $4s$ before $3d$ but are **removed from $4s$ first**. Fe^{2+} is $3d^6$, not $3d^44s^2$.

Rules, not formulas — recall these as facts

Ask	Answer
Highest oxidation state in 3d series	+7 in Mn (as KMnO_4); Cr reaches +6
Most stable oxidation state across the series	+2 for most, becoming more stable to the right
Why coloured	d-d transition of an unpaired d electron; Sc^{3+} (d^0) and Zn^{2+} (d^{10}) are colourless
Lanthanoid contraction	Steady size decrease across 4f, from poor shielding by f electrons
Its main consequence	Zr and Hf (also Nb/Ta) have almost identical radii and are hard to separate
Why alloys form readily	Similar atomic radii let one metal substitute for another in the lattice
Why good catalysts	Variable oxidation states and the ability to adsorb reactants on the surface
Most common lanthanoid oxidation state	+3; actinoids show far more variability

CH 5 Coordination Compounds — 6 formulas · 11 marks

● C5.1 Oxidation number of the metal inside a complex

$$x + \sum (\text{ligand charges}) = \text{charge on the coordination sphere}$$

Symbol	Meaning	Unit
x	oxidation number of the central metal	dimensionless
neutral	NH_3 , H_2O , CO , en — contribute 0	dimensionless
anionic	Cl^- , CN^- , OH^- contribute -1 ; $\text{C}_2\text{O}_4^{2-}$ contributes -2	dimensionless

Use it when:

Always — it is the first step of every naming, VBT and CFT question. In $\text{K}_4[\text{Fe}(\text{CN})_6]$: $x + 6(-1) = -4$, so $x = +2$.

Trap:

In $[\text{Ni}(\text{CO})_4]$ the ligands are neutral and the sphere is neutral, so nickel is in oxidation state **zero**. This exact complex is the standing trap.

● C5.2 Coordination number when a ligand grips more than once

$$\text{CN} = \sum_{\text{ligands}} (\text{denticity}) = \text{number of donor atoms bonded to the metal}$$

Symbol	Meaning	Unit
CN	coordination number	dimensionless
en	ethane-1,2-diamine — didentate, counts 2	dimensionless
$\text{C}_2\text{O}_4^{2-}$	oxalate — didentate, counts 2	dimensionless
EDTA	hexadentate, counts 6	dimensionless

Use it when:

Working out geometry. $[\text{Co}(\text{en})_3]^{3+}$ has three ligands but a coordination number of **6**.

Trap:

Count **donor atoms**, not ligands. This is the difference between getting octahedral and getting the wrong geometry entirely.

● C5.3 Tetrahedral splitting compared with octahedral

$$\Delta_t = \frac{4}{9} \Delta_o$$

Symbol	Meaning	Unit
Δ_o	octahedral crystal field splitting energy	J mol ⁻¹ (or cm ⁻¹)
Δ_t	tetrahedral splitting energy	J mol ⁻¹ (or cm ⁻¹)

Use it when:

Asked to relate the two, or to explain why tetrahedral complexes are essentially always high spin.

Trap:

Because Δ_t is small, it is almost never bigger than the pairing energy — so **no low-spin tetrahedral complexes** in this syllabus. Also note the splitting is *inverted*: *e* below *t₂*.

○ **C5.4 Crystal field stabilisation energy of an octahedral complex**

$$\text{CFSE} = (-0.4x + 0.6y) \Delta_o \quad \text{for } t_{2g}^x e_g^y$$

Symbol	Meaning	Unit
x	electrons in the lower t_{2g} set	dimensionless
y	electrons in the upper e_g set	dimensionless
CFSE	crystal field stabilisation energy	J mol ⁻¹ (or units of Δ_o)

Use it when:

Comparing the stability of two configurations. Each t_{2g} electron is stabilised by $0.4\Delta_o$; each e_g electron is destabilised by $0.6\Delta_o$.

Trap:

The barycentre must balance: $3 \times 0.4 = 2 \times 0.6$. If your two numbers do not satisfy that, you have the sets the wrong way round.

● **C5.5 High spin or low spin — deciding which**

$$\Delta_o > P \Rightarrow \text{low spin (pair up)} \quad \Delta_o < P \Rightarrow \text{high spin (stay unpaired)}$$

Symbol	Meaning	Unit
Δ_o	crystal field splitting energy	J mol ⁻¹
P	pairing energy	J mol ⁻¹
strong field	CN ⁻ , CO, NH ₃ — large Δ_o , low spin	—
weak field	I ⁻ , Br ⁻ , Cl ⁻ , F ⁻ , H ₂ O — small Δ_o , high spin	—

Use it when:

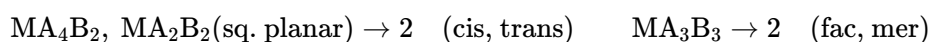
Predicting magnetic behaviour or hybridisation. Only matters for d^4 to d^7 ; below and above that there is no choice to make.

Trap:

Spectrochemical series, weakest to strongest:



○ **C5.6 How many geometrical isomers a complex has**



Symbol	Meaning	Unit
cis	identical ligands adjacent (90°)	—
trans	identical ligands opposite (180°)	—
fac	three identical ligands on one triangular face	—
mer	three identical ligands around a meridian	—

Use it when:

"What type of isomerism is shown by..." — check the formula shape first, then the geometry.

Trap:

Tetrahedral complexes show no geometrical isomerism — every position is adjacent to every other, so there is no "trans". Only the **cis** form of $[M(en)_2X_2]$ is optically active.

CH 6 Haloalkanes and Haloarenes — 2 formulas + orders · 6 marks

An organic chapter, so almost everything is a reactivity order rather than an equation. The two computable things come first; the orders that actually earn the marks are collected below them.

● **C6.1 The rate law that identifies the substitution mechanism**



Symbol	Meaning	Unit
S_N2	second order, one step, transition state, inversion	k in $L \text{ mol}^{-1} \text{ s}^{-1}$
S_N1	first order, two steps, carbocation, racemisation	k in s^{-1}
$[RX]$	concentration of the alkyl halide	mol L^{-1}

Use it when:

Asked for two differences between the mechanisms, or told how the rate responds to changing the nucleophile's concentration.

Trap:

The reactivity orders are **opposite**: S_N2 runs $1^\circ > 2^\circ > 3^\circ$ (steric hindrance), S_N1 runs $3^\circ > 2^\circ > 1^\circ$ (carbocation stability). Check which mechanism the question named before you answer.

○ **C6.2 Degree of unsaturation from a molecular formula**

$$\text{DoU} = \frac{2C + 2 + N - H - X}{2}$$

Symbol	Meaning	Unit
C, H, N	counts of carbon, hydrogen, nitrogen	dimensionless
X	count of halogen atoms	dimensionless
DoU	rings plus π bonds	dimensionless

Use it when:

A molecular formula is given and you must decide how many isomers or what skeleton is possible. Oxygen is ignored — it does not change the count.

Trap:

Halogen counts like hydrogen (subtract it). A benzene ring alone gives $\text{DoU} = 4$: three π bonds plus one ring.

The orders that earn the marks

Comparison	Order, and the reason
Reactivity of RX to substitution	$\text{RI} > \text{RBr} > \text{RCl} > \text{RF}$ — C-X bond weakens down the group
S_N1 rate	$3^\circ > 2^\circ > 1^\circ$ — carbocation stability; benzylic and allylic beat all three
S_N2 rate	$\text{CH}_3\text{X} > 1^\circ > 2^\circ > 3^\circ$ — steric hindrance
Alcohol + HX	$3^\circ > 2^\circ > 1^\circ$, and $\text{HI} > \text{HBr} > \text{HCl}$
Boiling point of isomers	Falls as branching rises — less surface contact, weaker van der Waals
Melting point of dihalobenzenes	para > ortho $\approx$ meta — para packs better in the crystal lattice
Free radical stability	benzylic $\approx$ allylic > $3^\circ > 2^\circ > 1^\circ$ — resonance beats hyperconjugation
Aqueous vs alcoholic KOH	Aqueous $\rightarrow$ substitution (alcohol); alcoholic $\rightarrow$ elimination (alkene)
Markovnikov vs peroxide	Peroxide effect works with HBr only
Saytzeff (NCERT: Zaitsev)	The more substituted alkene is the major elimination product

CONST Constants and conversions — *know the units too*

● K Every constant this paper can hand you, with its unit

Symbol	Meaning	Value and unit
R	gas constant, energy form	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
R	gas constant, pressure-volume form	$0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$
F	Faraday constant	96487 C mol^{-1}
K_b	ebullioscopic constant, water	$0.52 \text{ K kg mol}^{-1}$
K_f	cryoscopic constant, water	$1.86 \text{ K kg mol}^{-1}$
N_A	Avogadro constant	$6.022 \times 10^{23} \text{ mol}^{-1}$
2.303	converts $\ln$ to $\log_{10}$	dimensionless
0.693	$\ln 2$, for first-order half-life	dimensionless
0.059	$2.303RT/F$ at 298 K, for Nernst	V
T	0 °C in kelvin	273.15 K
STP	molar volume of an ideal gas	22.4 L mol^{-1}

Trap:

Match R to the units you want out. Using 8.314 with pressures in atm, or 0.0821 when the answer should be in joules, is a whole-question error and not a rounding one.

Built from Sourabh Raina's one-shot and PYQ videos for Chapters 1-6, and cross-checked against NCERT Class XII Chemistry (Rationalised 2022-23). NCERT conventions used throughout: **0.059** rather than 0.0591 in the Nernst equation, $F = 96487 \text{ C mol}^{-1}$, molar conductivity in $\text{S cm}^2 \text{ mol}^{-1}$, and **Zaitsev** (which NCERT notes is "also pronounced as Saytzeff").

The six chapter pages carry the teaching and the past-year questions; this page is only the recall layer. Derivations for the formulas that have them live on the companion page,

Chemistry, Derived.